

Karan Gupta

Software Engineer | Full Stack Developer | Jalandhar, Punjab, India

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Portfolio | GitHub | LinkedIn

Projects

EventFlow – Distributed Workflow Orchestration Engine

Tech: FastAPI, Python, PostgreSQL, Redis Streams, gRPC, Docker, WebSockets

GitHub: github.com/guptakaran20/EventFlow | Live: goeventflow.vercel.app

- Built an event-driven workflow orchestration engine with custom workers, graph-based routing, retries, and a Dead Letter Queue using Redis Streams.
- Engineered fault-tolerant worker execution with heartbeats, persistent job state, and automatic stuck-job recovery using Redis Streams `XPENDING/XCLAIM`.
- Developed real-time execution updates over WebSockets with PostgreSQL persistence, structured observability APIs, metrics, logs, and DLQ management.

CodeArena – Real-Time Coding Battle Platform

Tech: Next.js, Node.js, Express, MongoDB, Redis, Socket.IO, Docker, AWS EC2

GitHub: github.com/guptakaran20/CodeBattle | Live: codearenabattle.vercel.app

- Engineered a multiplayer competitive coding platform with real-time battle rooms, synchronized leaderboards, contest state, and live player presence.
- Built secure REST APIs with JWT/OAuth authentication and integrated Piston for isolated code execution, using Redis and Socket.IO for real-time contest synchronization.

ImportlyAI – AI-Powered CSV Import Platform

Tech: Next.js, TypeScript, Node.js, Express.js, Gemini API, PapaParse

GitHub: github.com/guptakaran20/ImportlyAI | Live: importlyai.vercel.app

- Engineered a multi-stage AI extraction pipeline using Gemini to map arbitrary CSV structures into a standardized CRM schema.
- Developed an automated benchmark suite with 180+ tests and optimized batched inference for large, inconsistent datasets.

SponsorGrid – Sponsorship Management Platform

Tech: Next.js, TypeScript, PostgreSQL, Prisma ORM

GitHub: github.com/guptakaran20/Sponsorship | Live: trysponsorgrid.vercel.app

- Developed a full-stack SaaS platform for managing sponsors, funding workflows, and sponsorship data using Next.js, PostgreSQL, Prisma ORM, and OAuth.

Technical Skills

Languages: C++, TypeScript, JavaScript, Python, SQL

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Backend: Node.js, Express.js, REST APIs, gRPC, FastAPI, Socket.IO, WebSockets, JWT Authentication, Google OAuth

Databases: PostgreSQL, MongoDB, MySQL, Redis, Prisma ORM

DevOps & Tools: Docker, Docker Compose, AWS EC2, Nginx, Git, GitHub, Vercel, CI/CD

Mobile: Capacitor, Firebase Cloud Messaging, Android

Experience

Full Stack Developer

Jun 2026 – Present

XCEED NIT Jalandhar

ILEED – Intelligent Learning Engagement and Entity Detection

- Architected and developed core backend services for an intelligent attendance platform serving **1,000+ students** across **30+ cameras**, integrating FAISS-based face recognition for automated ground-truth tracking.
- Engineered scalable RTSP stream-processing pipelines and backend APIs for automated attendance workflows, camera management, and real-time ground-truth acquisition.
- Built the cross-platform mobile application using **Capacitor**, with Firebase Cloud Messaging for real-time attendance alerts and native mobile integration.
- Automated ERP face-embedding synchronization through asynchronous ML processing, with background generation, progress monitoring, consistency checks, and automatic synchronization after photo updates.
- Improved reliability of live camera previews by implementing automatic retry and stream recovery mechanisms for intermittent RTSP failures.

Education

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar

B.Tech – Instrumentation and Control Engineering

2025 – 2029

CGPA: 8.49

Achievements

- Solved 500+ Data Structures and Algorithms problems; achieved a 1857 LeetCode rating, earning Knight status.